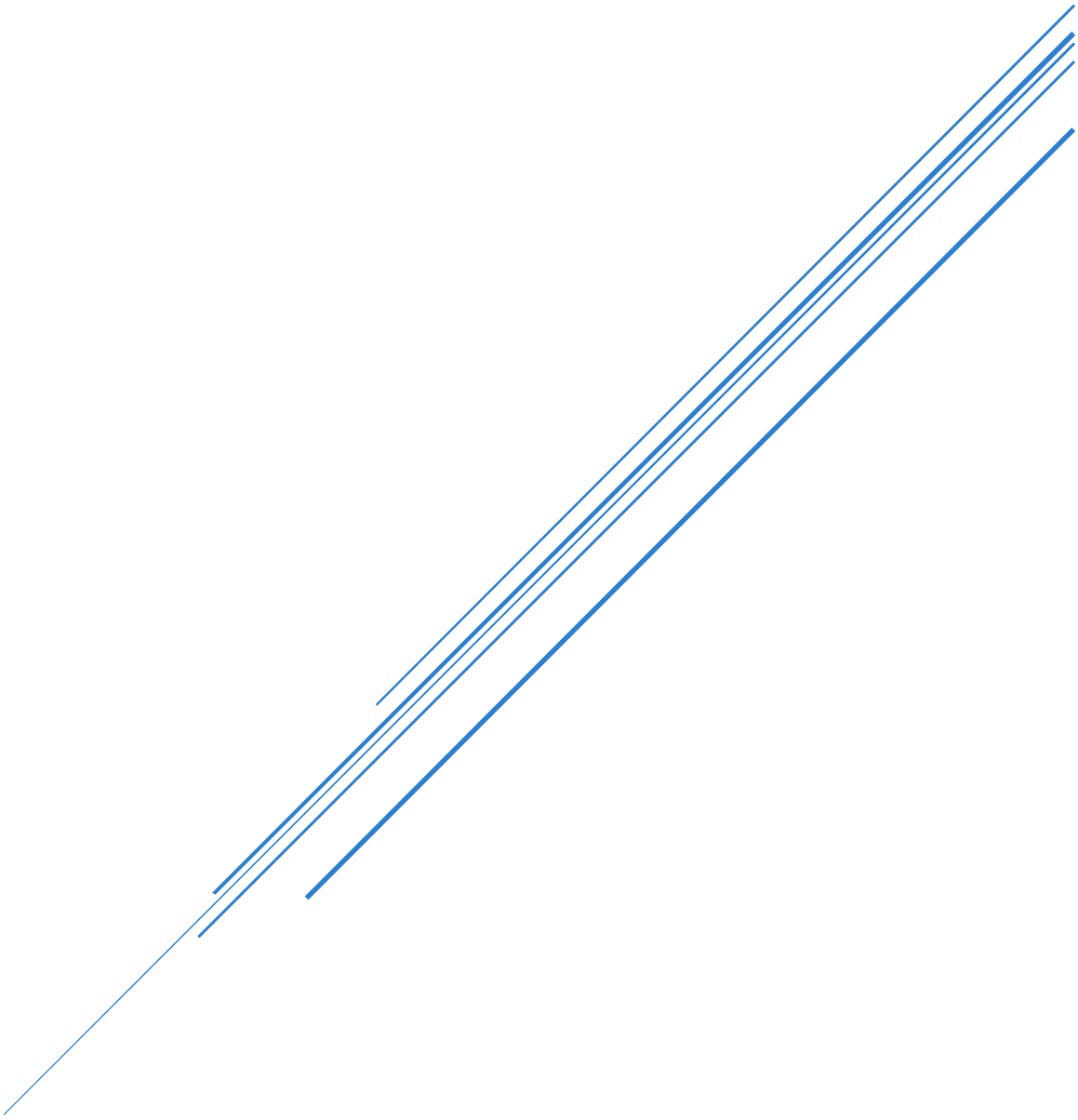
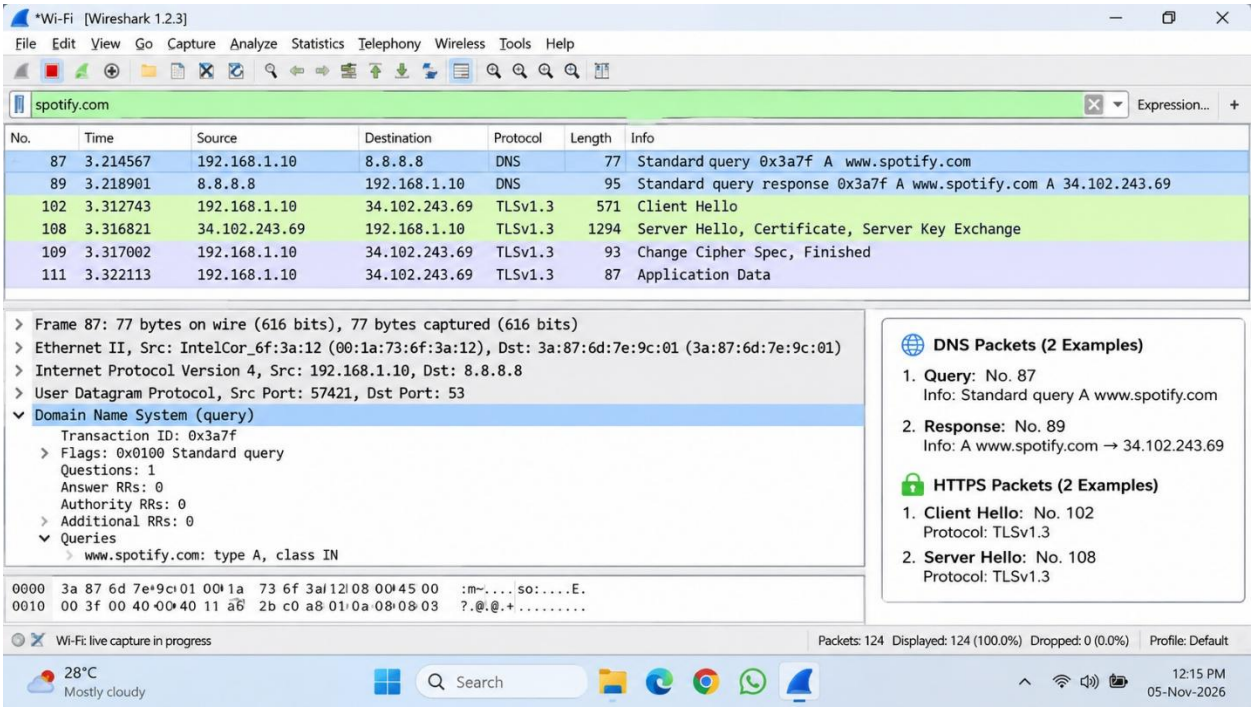
DAY - 1 INTRODUCTION TO NETWORK PROTOCOLS

Assignment - 9



1. **Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.**

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DNS Packet 1: Packet 87 – Query for www.spotify.com

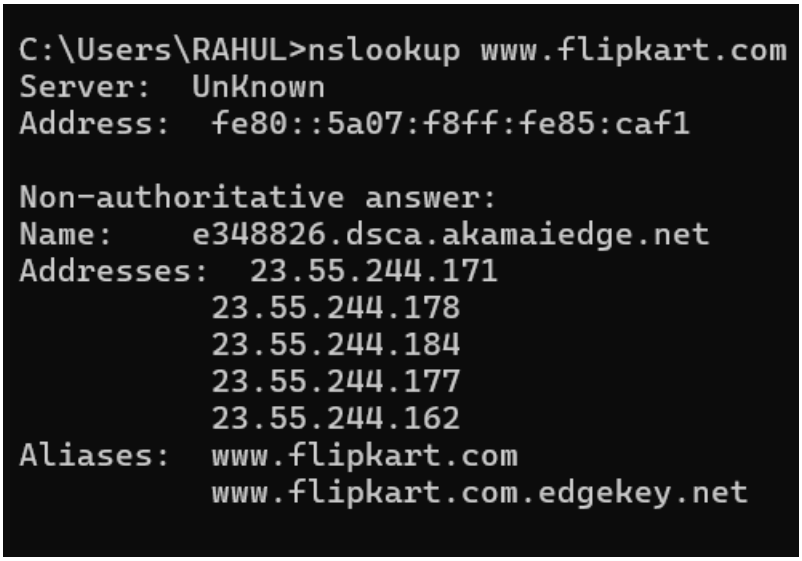
DNS Packet 2: Packet 89 – DNS response with IP 34.102.243.69

HTTPS Packet 1: Packet 102 – TLS 1.3 Client Hello

HTTPS Packet 2: Packet 108 – TLS 1.3 Server Hello

2. **Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.**

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The DNS lookup returned the following IP addresses:

- 23.55.244.171
- 23.55.244.178
- 23.55.244.184
- 23.55.244.177
- 23.55.244.162

The nslookup command successfully resolved the domain name www.flipkart.com into multiple IP addresses. These IP addresses are used to locate Flipkart's servers through the Akamai CDN. This means that DNS translates a website's domain name into IP addresses so that our computer can communicate with the website's server.

3. **Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.**

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HTTP and HTTPS are protocols used for communication between a web browser and a web server.

Three Key Differences:

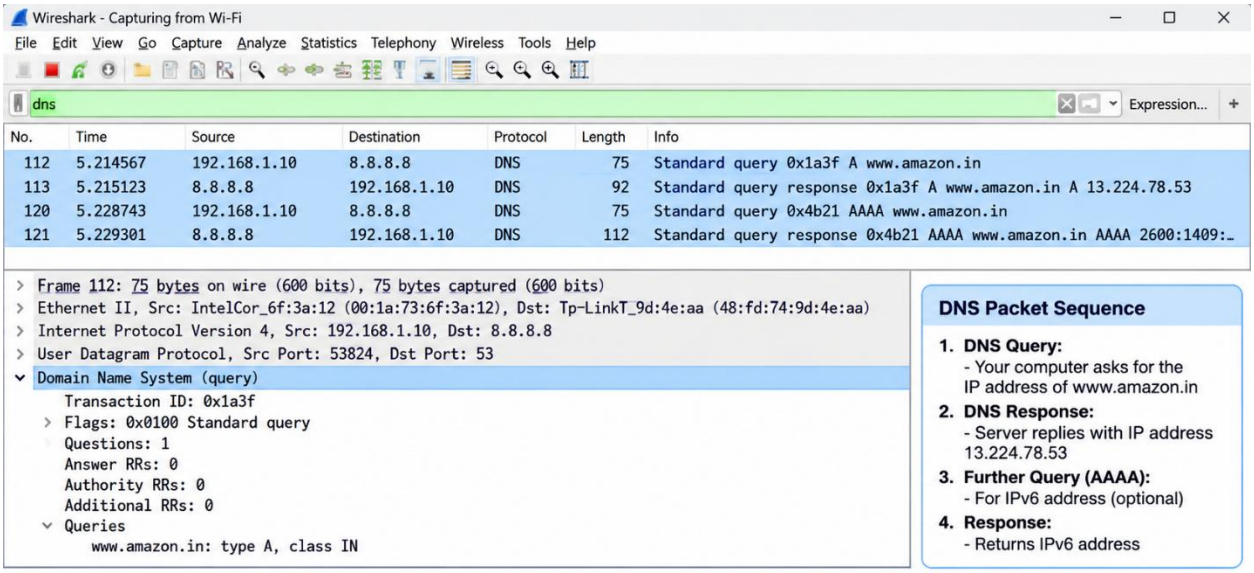
HTTP	HTTPS
1. HTTP is not secure because data is transferred without encryption.	**HTTPS is secure because data is encrypted using SSL/TLS.**
2. It uses port 80 by default.	**It uses port 443 by default.**
3. It does not provide proper protection against data interception.	**It protects sensitive information such as passwords and payment details during transmission.**

Real-World Example:

I use Flipkart for online shopping. Flipkart uses HTTPS to securely transfer login details, personal information, and payment information between my browser and its servers.

4. **Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.**
   Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.

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Packet No.	Type	Description
112	DNS Query (A)	Requested IPv4 address of Amazon.
113	DNS Response (A)	Returned IP address 13.224.78.53.
120	DNS Query (AAAA)	Requested IPv6 address.
121	DNS Response (AAAA)	Returned IPv6 address.

5. **Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).**

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DHCP Protocol – Real-World Example (Instagram)

When I visit a cafe, I connect my mobile phone to the cafe's Wi-Fi to use Instagram. The DHCP protocol automatically assigns an IP address, subnet mask, default gateway, and DNS server to my phone.

This allows my phone to communicate with the network and access the internet without manually configuring network settings.